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AI-Based Storage Tracking System Using R-CNN and RFID Integration

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Abstract

This project aims to develop an automated tracking system that uses object detection system using a Region-Based Convolutional Neural Network (R-CNN) to identify empty slots within a multiple partition. The model was trained on a custom dataset and integrated for real-time inference. Results show accurate classification and localization performance suitable for inventory automation.

Problem Statement

- Manual storage tracking is time-consuming and prone to human error.
- Existing systems struggle to accurately detect empty or partially filled slots.
- Inconsistent detection performance occurs under varying lighting or visual conditions.

Introduction

- Inventory monitoring systems are crucial for maintaining efficiency in warehouses and storage facilities.

Traditional methods depend on:

- Manual inspection (time-consuming and prone to human error).
- This study proposes using deep learning-based object detection with RCNN.
- The goal is to enable an automated and accurate identification of slot occupancy, thereby enhancing reliability and reducing the need for manual intervention.

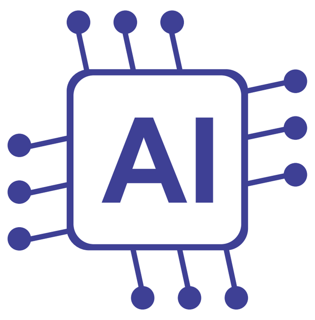
Objectives

1. To design and develop an automated storage tracking system
2. To implement object detection for identifying empty slots within a partition using a two-stage model (RCNN)
3. To achieve at least 95% detection accuracy in recognizing and counting items within each slot of the partition.

Methodology

- The dataset was collected and labeled using LabelImg,
- The images are augmented and compress to improve model robustness.
- A Region-based CNN was implemented using PyTorch, trained with optimized hyperparameters for best detection accuracy:

BATCH_SIZE = 1
NUM_EPOCHS = 150
LEARNING_RATE= 0.0005
MOMENTUM = 0.9
WEIGHT_DECAY = 0.0005
SCHEDULER_PATIENCE = 3
MIN_LR = 1e-5



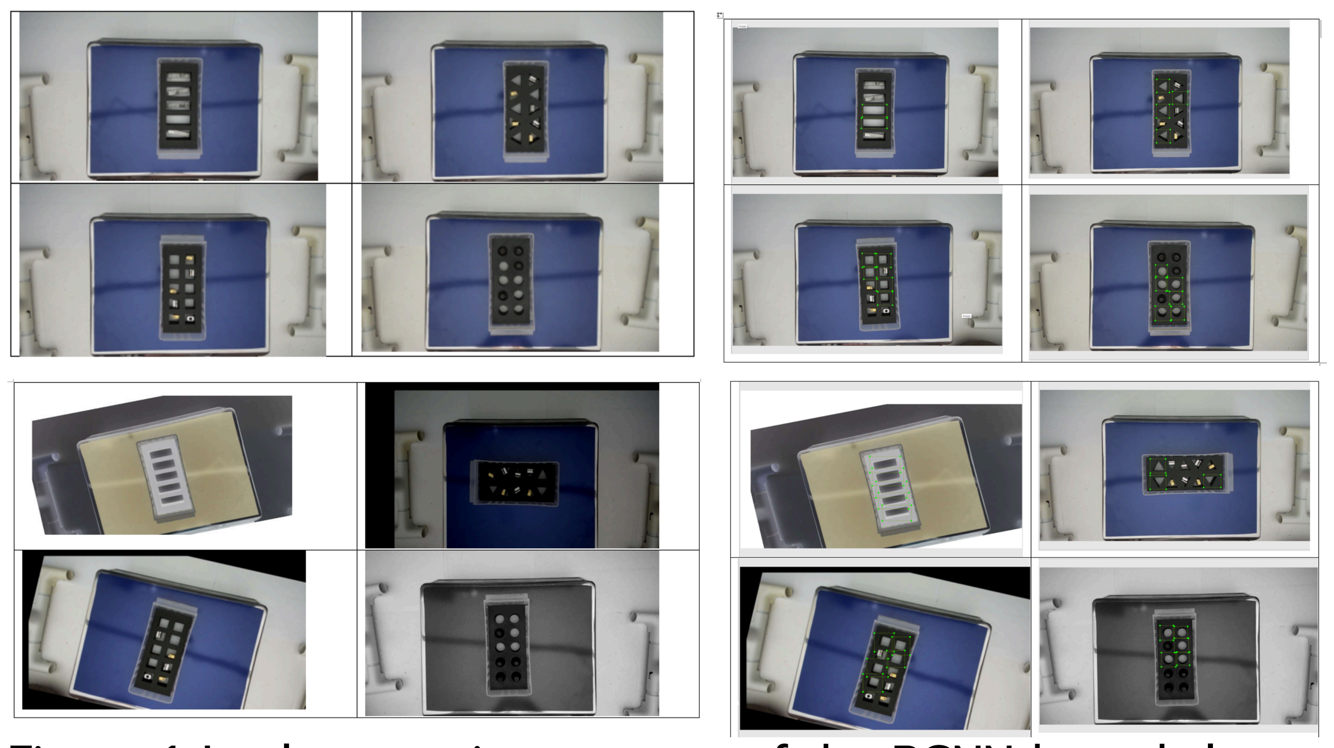


Figure 1. Implementation process of the RCNN-based slot detection system.

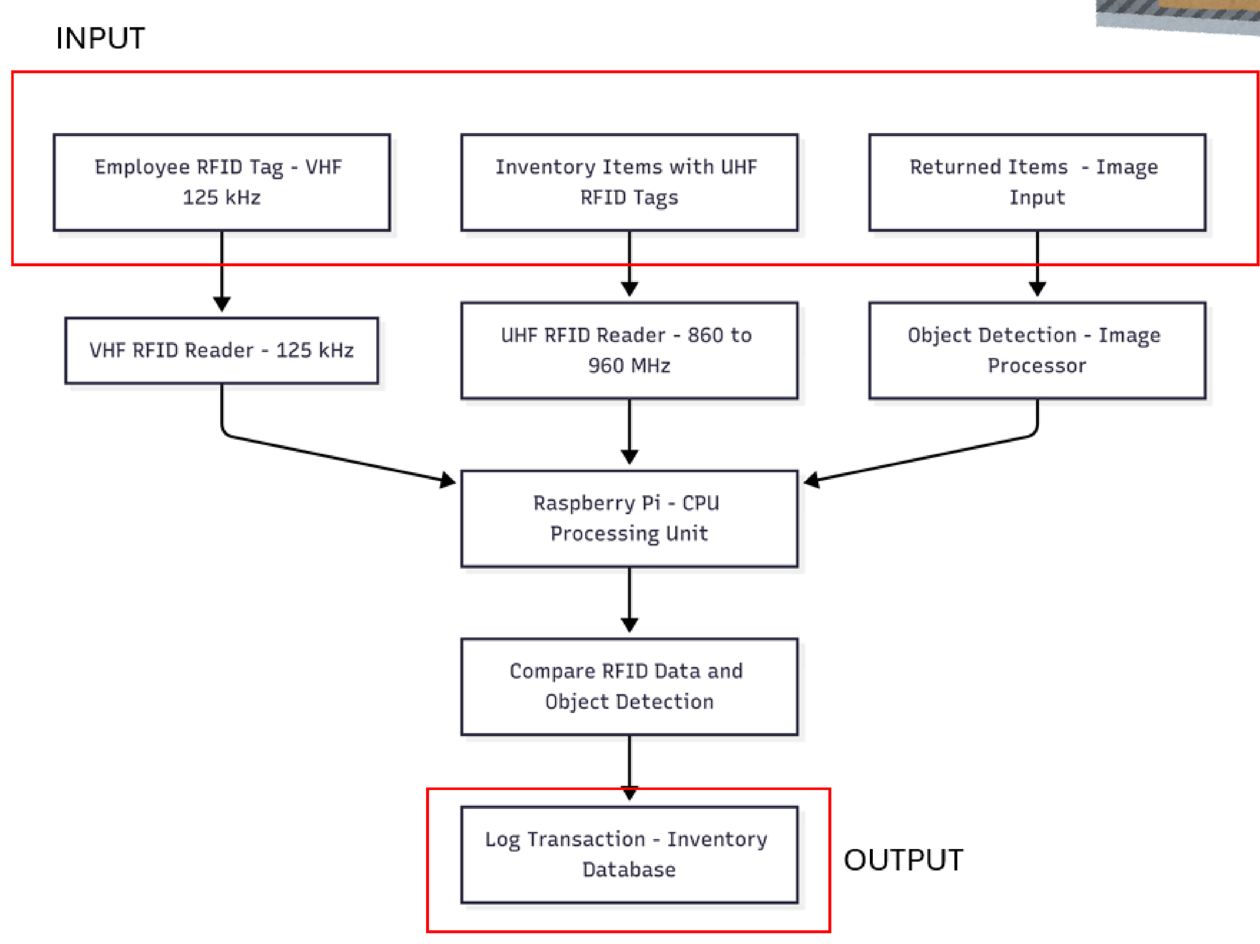


Figure 2 Flowchart of the Methodology

Results

The trained RCNN model successfully detected and classified empty slot occupancy with high accuracy. Evaluation metrics, including confusion matrix and precision-recall analysis, confirmed effective learning performance. Visual inference results demonstrated clear differentiation between empty and filled slots under varied lighting and angles.

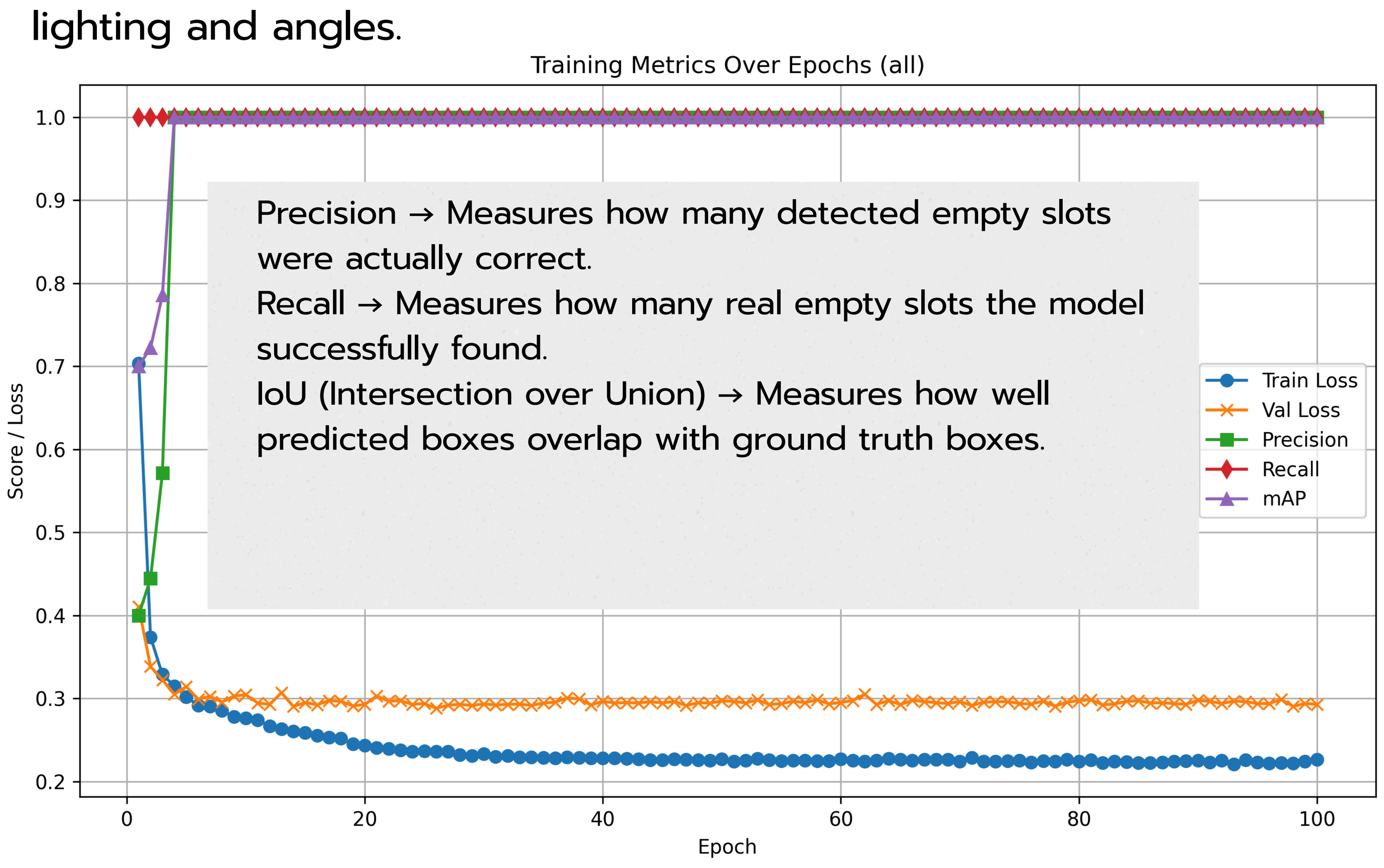


Figure 4 Model Training Metrics

True Positive ←[2340 45] → False Positive
False Negative ←[0 0] → True Negative

- Precision: 0.9811, Recall: 1.0000, F1: 0.9905, Average IoU: 0.8402
- High precision (98%) → Few false alarms.
- Perfect recall (100%) → No missed empty slots.
- Strong IoU (84%) → Bounding boxes fit accurately.

Figure 5 Successfully detected the empty slots in the partitions.

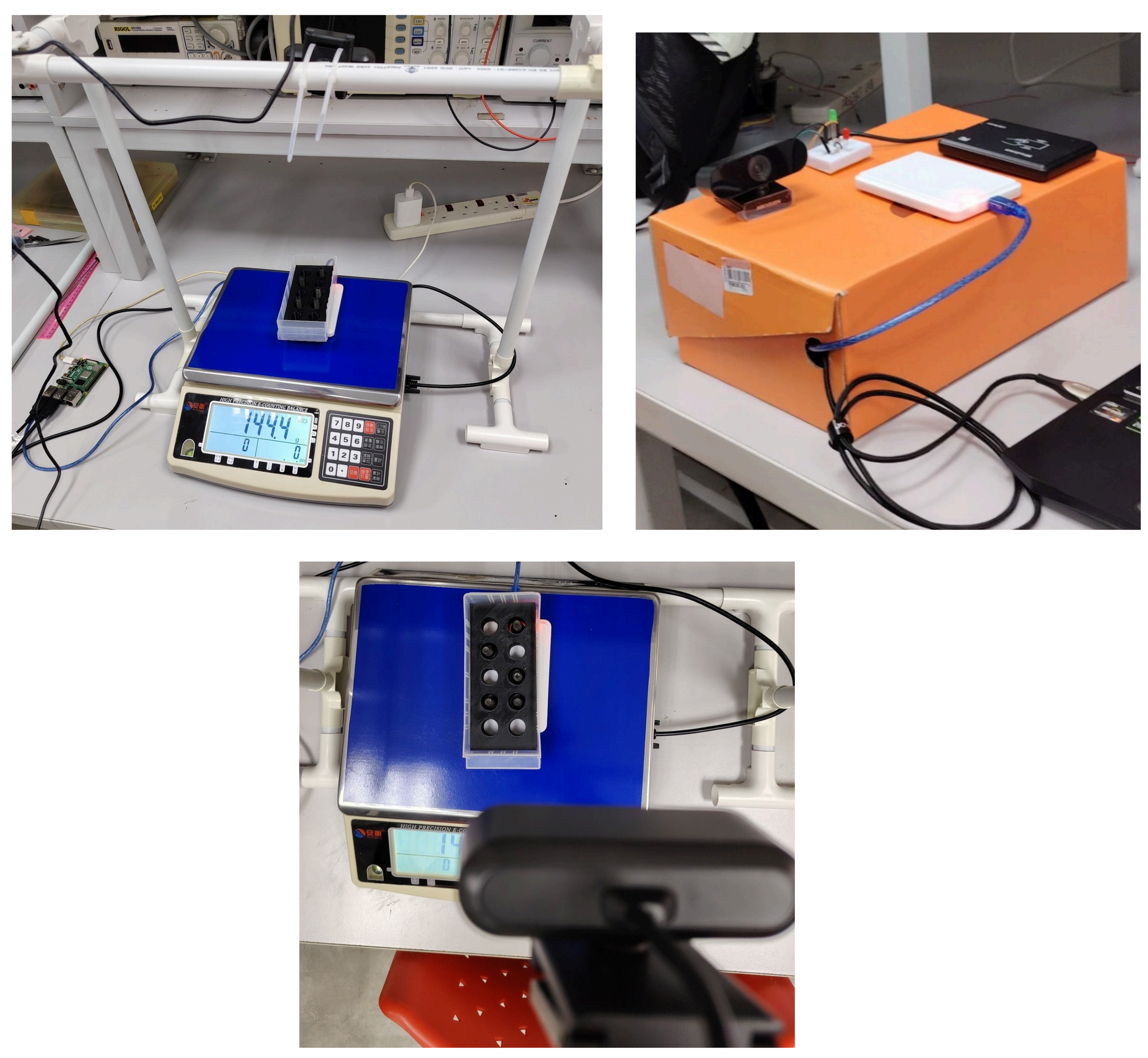


Figure 6 Setup of the project

Acknowledgements

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References

RCNN - https://blog.roboflow.com/what-is-r-cnn
Confusion Matrix - https://www.v7labs.com/blog/confusion-matrix-guide

Conclusion

This research highlights the effectiveness of RCNN in automating tracking system. Achieving its objective offering accurate visual recognition, reducing manual workload and increasing operational efficiency. Future work may include real-time deployment and integration with Power Apps for live monitoring and decision support.